

B. Tech. Project Mid-Semester Evaluation

Explainable Deep Learning Framework for Automated ECG Arrhythmia Detection and Classification

Presented by

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Introduction & Motivation

Background

- The ECG is a widely used non-invasive tool for detecting cardiac arrhythmias; automating its interpretation is an active research area.
- Deep learning dominates: CNNs are used in 58.7% of 368 studies [1] and 62% of 119 studies [2]; hybrid CNN-recurrent and Transformer models are increasingly common.

Motivation

Accuracy that does not transfer

Reported accuracies above 98% fall considerably on unseen patients, external databases or noisy signals [1], [2].

Explanations that are not checked

Explanations attached to such models are rarely validated against clinical ECG knowledge [3].

Clinical use needs a classifier that works on new patients, explains its decisions, and stays reliable on real, noisy recordings.

Problem Statement

There is a need for an ECG arrhythmia classification framework whose performance is measured on unseen patients, whose explanations are checked for consistency and clinical plausibility, and whose predictions and explanations are evaluated for robustness under realistic noise.

Existing limitations

Evaluation protocol

F1 fell from 95.52% (intra-patient) to 83.89% (inter-patient) in studies reporting both [1].

Black-box models

Fewer than 10% of studies validated explanation maps against clinically meaningful ECG structures [2].

Narrow robustness tests

Noise robustness is often tested with additive white Gaussian noise only [4].

Why it matters: a model that is only tested on familiar patients and clean signals, with unchecked explanations, cannot be trusted on new, real recordings.

Objectives

- 01** **Objective 1: Patient-independent baseline**

Develop a hybrid deep learning arrhythmia classifier evaluated under a strict inter-patient protocol (MIT-BIH DS1/DS2) with external testing on a second database; report per-class and macro F1 alongside accuracy.
- 02** **Objective 2: Validated explanations**

Apply post-hoc XAI methods (SHAP and Grad-CAM) and measure the within-class consistency of explanations and their agreement with ECG features (P wave, QRS complex, RR interval).
- 03** **Objective 3: Robustness of predictions and explanations**

Assess how classification performance and explanation stability degrade under realistic NSTDB noise at graded SNRs, using uncertainty estimates to flag unreliable predictions.

Literature Review

36 papers surveyed (5 reviews, 31 original studies). Most relevant studies:

Ref.	Dataset(s)	Model	XAI	Split	Key result
[5]	MIT-BIH + LTAF	Lightweight CNN-LSTM	SHAP	Not stated (85/15)	Acc 98.24%, Se 86.1%; 0.16 MB
[3]	MIT-BIH + 12-lead	1D CNN	4 SHAP variants	Not stated (80/20)	Val acc 98.30%; 73.45% on combined data
[4]	MIT-BIH; INCART, SVDB, PTB	CNN-BiLSTM + CapsNet fusion	Branch weights	Inter-patient	Acc 99.55%; F1 32.46% at 5 dB AWGN
[6]	MIT-BIH	CNN-BiLSTM + cGAN	Not stated	Both	99.00% beat-wise vs 91.69% patient-wise

Key findings

- Architectures: CNNs dominate; hybrids add a recurrent or attention module (CNN-LSTM [5], CNN-BiLSTM [6]); Transformers gain 0.845% over 1D-CNNs at higher cost [2].
- XAI: SHAP is the most used method, then Grad-CAM; APB/PVC explanations were inconsistent within a class [3].
- Evaluation: MIT-BIH dominates (61% of studies [1]); external validation in only 18.5% of studies, with a mean accuracy drop of 8.7% [2].

Research Gap

Gap 1 – Explanations rarely validated

< 10% of studies validate maps against ECG structures [2]; explanations inconsistent within a class [3].

→ **Addressed by** Objective 2: region-share and deletion checks

Gap 2 – XAI seldom patient-independent

XAI studies often do not describe patient separation, though inter-patient evaluation lowers performance [1], [6].

→ **Addressed by** Objectives 1 + 2: XAI on a DS1/DS2 model

Gap 3 – Noise tests narrow, not linked to XAI

Noise tests limited to AWGN [4] or absent; no surveyed paper tests whether explanations stay stable under noise.

→ **Addressed by** Objective 3: NSTDB noise + heatmap stability

Gap 4 – Weak minority classes

Patient-wise Fusion and S recall only 0.021 and 0.209 [6]; little external validation [2].

→ **Addressed by** Objective 1: per-class / macro F1, external test

Dataset

Framework datasets

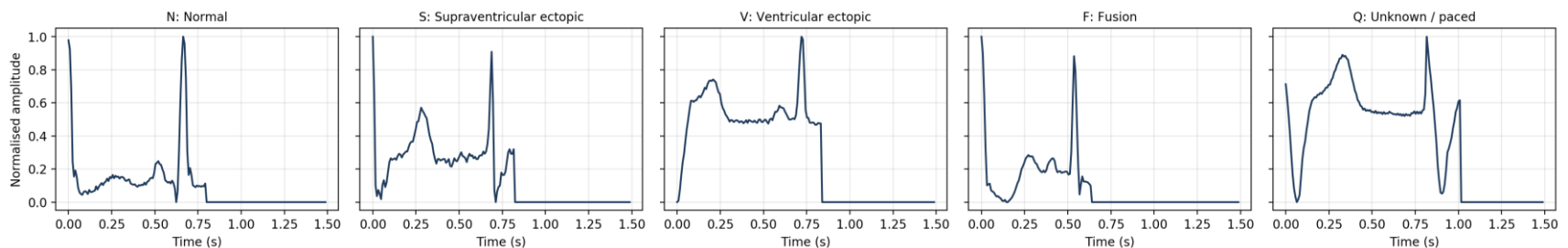
- MIT-BIH Arrhythmia Database: AAMI classes, inter-patient DS1/DS2 split.
- External test: INCART (preferred; stretch goal).
- MIT-BIH NSTDB: baseline wander, muscle artefact, electrode motion noise.

Used so far (preliminary)

- Pre-segmented MIT-BIH heartbeat dataset [7].
- 187-sample beats (1.5 s at 125 Hz), scaled to [0, 1], zero-padded.
- 5 AAMI classes: N, S, V, F, Q.

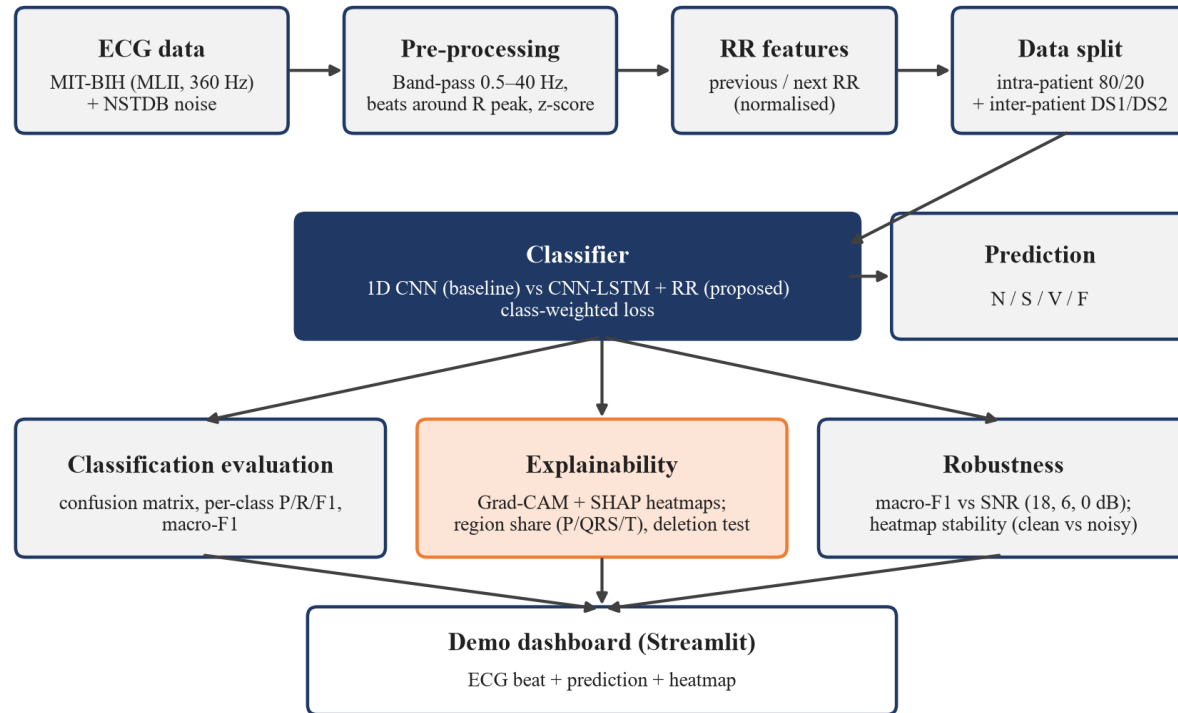
File	N	S	V	F	Q	Total
Train file	72,471 (82.77%)	2,223 (2.54%)	5,788 (6.61%)	641 (0.73%)	6,431 (7.35%)	87,554
Test file	18,118 (82.76%)	556 (2.54%)	1,448 (6.61%)	162 (0.74%)	1,608 (7.35%)	21,892

Split: train file 80/20 (stratified, seed 42) → 70,043 train / 17,511 validation; test file (21,892) used only for final evaluation. Split is by beat → intra-patient.



One example beat per class: starts at the R peak, second peak ≈ next R peak, then zero padding

Proposed Methodology



Block diagram of the proposed framework

Pre-processing: filtering, R-peak-centred beats, normalisation, RR features.

Split: intra-patient 80/20 + inter-patient DS1/DS2.

Explain: Grad-CAM [8] + Gradient SHAP [9].

Robustness: NSTDB 'ma' noise at 18 / 6 / 0 dB SNR; macro F1 vs SNR.

Model & Algorithms

Baseline 1D CNN (implemented): input 187×1 beat $\rightarrow$ softmax over 5 classes

Layer	Output shape	Parameters
Conv1D 32, k = 7 $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool 2	(93, 32)	256 + 128
Conv1D 64, k = 5 $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool 2	(46, 64)	10,304 + 256
Conv1D 128, k = 3 $\rightarrow$ BN $\rightarrow$ ReLU (Grad-CAM layer) $\rightarrow$ MaxPool 2	(23, 128)	24,704 + 512
Global average pooling $\rightarrow$ Dense 64, ReLU $\rightarrow$ Dropout 0.3 $\rightarrow$ Dense 5, softmax	(5)	8,256 + 325
Total		44,741

Residual CNN (V2 final)

Conv 32 (k 7) stem + 3 residual blocks (32, 64, 128); each: two Conv1D (k 5) + BN with a shortcut (1 $\times$ 1 conv when channels change). 185,477 parameters.

Proposed: CNN-LSTM + RR

Same conv front end $\rightarrow$ LSTM (64); last hidden state + previous / next RR ($\div$ record mean RR) $\rightarrow$ Dense 64 $\rightarrow$ softmax over N, S, V, F.

Explainability

Grad-CAM [8] on the last conv activation (46 steps $\rightarrow$ 187 samples). Gradient SHAP [9] planned. Measures: region share, deletion test, stability.

Key design choices: square-root class weights (F:N 10.6:1 instead of 113:1); best epoch chosen on validation macro F1; per-class probability thresholds tuned on validation.

Experimental Setup

ID	Change added (cumulative)	Network	LR	Clip norm	BN mom.	Class weights	Augment.
E1	Epoch chosen on val. macro F1	CNN	1×10^{-3}	–	0.99	Balanced	No
E2	+ lower LR, clipping, BN 0.9	CNN	5×10^{-4}	1.0	0.9	Balanced	No
E3	+ square-root class weights	CNN	5×10^{-4}	1.0	0.9	Square-root	No
E4	+ light augmentation	CNN	5×10^{-4}	1.0	0.9	Square-root	Yes
E5	+ residual network	Residual CNN	5×10^{-4}	1.0	0.9	Square-root	Yes

B0 = V1 baseline (LR 10^{-3} , balanced weights, early stopping on validation loss, patience 6). Final = E5 + thresholds tuned on validation ($N \times 3$).

Training

Adam, batch 128, ≤ 30 epochs

Best weights on val. macro F1; LR halved after 3 non-improving epochs (min 10^{-5}); stop after 8

Augmentation: amplitude $\pm 10\%$, noise σ 0.01, shift ± 3 samples

Seed 42; test set never used for selection

Evaluation metrics

Accuracy

Per-class precision, recall, F1

Macro F1 – main metric ($N \approx 83\%$ of beats, so accuracy hides rare-class errors)

Confusion matrix

Tools & hardware

Python 3, TensorFlow 2.20 / Keras

NumPy, pandas, scikit-learn, Matplotlib, seaborn

wfdb, SHAP, Streamlit (next phase)

Kaggle notebooks, NVIDIA Tesla T4 GPU; 5 runs in 8.1 min

Work Done Till Now

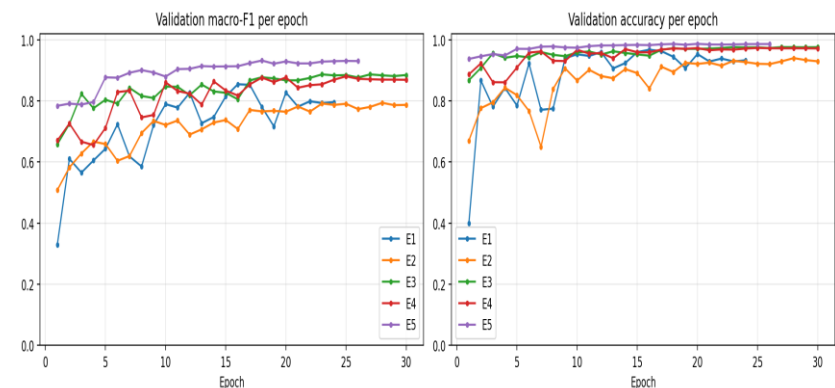
Completed milestones

- 1 Literature survey of 36 papers; research gaps identified
- 2 Framework design: AAMI classes, DS1/DS2 protocol, XAI metrics, noise settings
- 3 Data preparation: stratified 80/20 split, class weights
- 4 V1: baseline 1D CNN trained, evaluated, Grad-CAM per class
- 5 V2: 5 controlled experiments (E1–E5) + threshold tuning
- 6 Final model evaluated; Grad-CAM on the final model

Experiments conducted (test set, %)

ID	Change	Acc.	Macro F1
B0	V1 baseline	94.41	76.29
E1	Epoch on val. macro F1	96.28	83.96
E2	+ LR, clipping, BN	93.58	78.30
E3	+ sqrt class weights	97.39	87.58
E4	+ augmentation	97.02	86.20
E5	+ residual network	98.29	90.77
Final	E5 + tuned thresholds	98.41	91.10

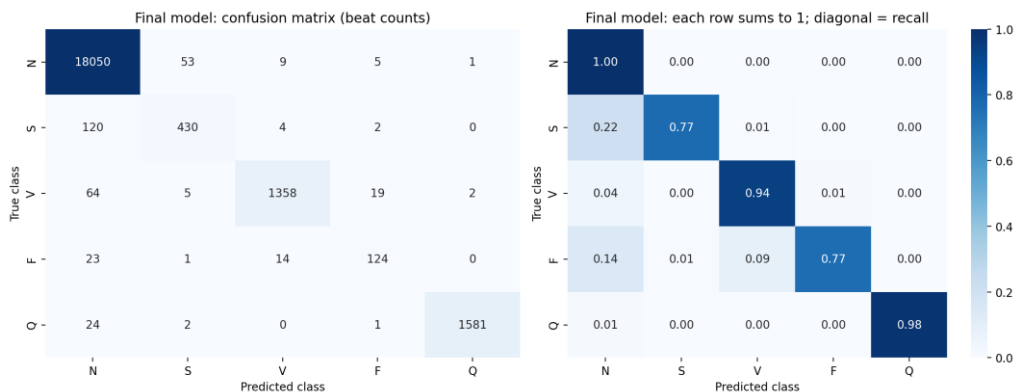
Same data, split and seed for every step; one change added at a time.



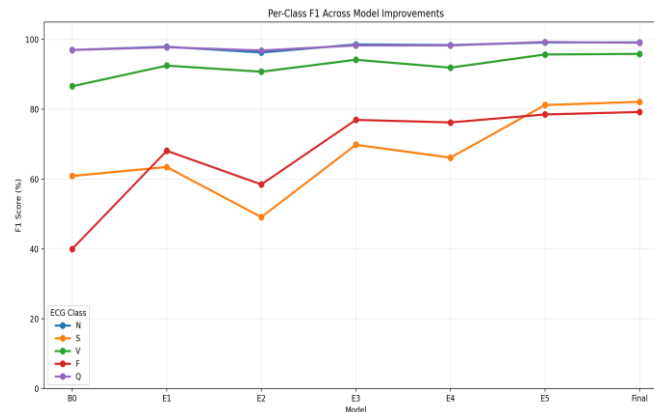
Validation macro F1 / accuracy per epoch, E1–E5

Preliminary Results

Accuracy 94.41% → 98.41% | Macro F1 76.29% → 91.10% (V1 baseline → V2 final)



Final model confusion matrix (counts; row-normalised)



Per-class F1, B0 → Final

Metric	V1 baseline	V2 final
Precision / Recall S (%)	55.86 / 66.91	87.58 / 77.34
Precision / Recall F (%)	26.02 / 86.42	82.12 / 76.54
Beats wrongly predicted as F	398	27
Beats wrongly predicted as S	294	61
S beats predicted as N	163 (29%)	120 (22%)
Parameters	44,741	185,477

Class	Prec.	Rec.	F1
N	0.987	0.996	0.992
S	0.876	0.773	0.821
V	0.981	0.938	0.959
F	0.821	0.765	0.792
Q	0.998	0.983	0.991
Macro	0.933	0.891	0.911

Test set: 21,892 beats; intra-patient split. F1 improved for every class (S 60.9 → 82.1, V 86.6 → 95.9, F 40.0 → 79.2).

Challenges & Key Findings

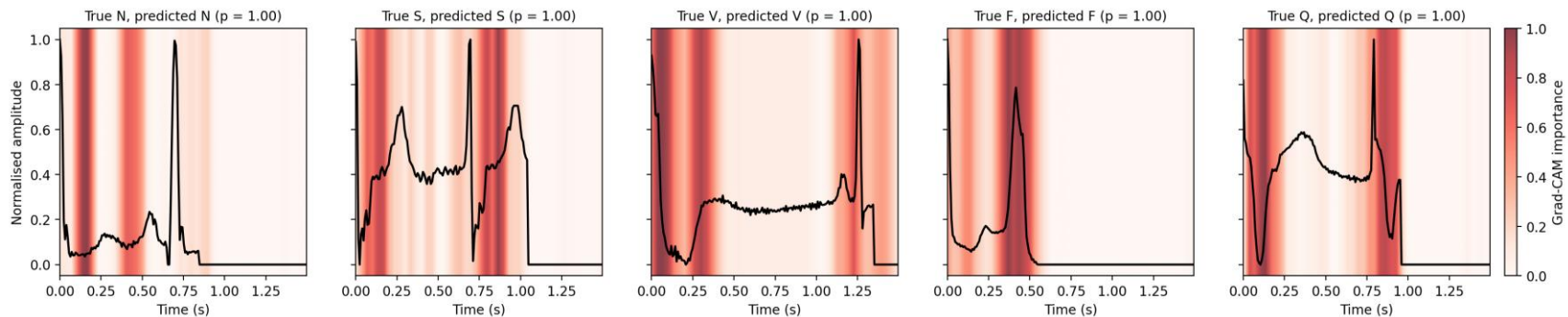
Challenges faced

- Class imbalance: balanced weights (F:N 113:1) gave F precision 26.0%.
- Unstable training: V1 validation accuracy swung 14.5% $\leftrightarrow$ 95.1%.
- E2 stability settings (-5.66) and augmentation E4 (-1.38) hurt; steps cumulative.
- Optimistic setting: pre-processed, intra-patient, 5 classes, one seed per run.

Key findings

- Biggest gains: square-root weights (E3, +9.28), epoch selection on macro F1 (E1, +7.67), residual network (E5, +4.57).
- F recall fell 86.4% $\rightarrow$ 76.5%: V1's high recall came from over-predicting F.
- Largest remaining error S $\rightarrow$ N (22%): S differs from N mainly in timing $\rightarrow$ RR features.

Explainability finding



Grad-CAM of the final model (most confident correct beat per class): heat often near the second R peak $\rightarrow$ possible implicit RR cue; qualitative only so far

Future Work & Expected Outcomes

Remaining tasks

- 1 Port pipeline to raw MIT-BIH; pre-processing + DS1/DS2 split
- 2 Train hybrid CNN-LSTM + RR; per-class and macro F1; external test
- 3 Apply Grad-CAM + SHAP; region-share, deletion and consistency checks
- 4 NSTDB noise at graded SNRs; explanation stability and uncertainty
- 5 Analyse results; final report and presentation

Deliverables

- Patient-independent classifier with per-class and macro results (inter-patient + external).
- Quantitative evaluation of explanations with two XAI methods.
- Robustness and uncertainty analysis under NSTDB noise.
- Final report and presentation.

Expected results

- Intra-patient macro F1 $\approx$ 90%; DS1/DS2 clearly lower [1], [6]; RR features should raise S recall.
- V beats: larger QRS importance share than N; top-sample deletion $\gg$ random deletion.
- Macro F1 and explanation stability fall as SNR drops 18 $\rightarrow$ 0 dB.

Anticipated challenges

- Inter-patient scores clearly lower; S and F remain hardest.
- External testing and MC dropout only if time permits.

Conclusion & References

Conclusion

- A working training, evaluation and Grad-CAM pipeline is in place.
- Macro F1 **76.29% → 91.10%** and accuracy **94.41% → 98.41%** via epoch selection, square-root weights and a residual CNN.
- Results are intra-patient: a development baseline for the patient-independent framework.
- **Next:** raw MIT-BIH + DS1/DS2 → CNN-LSTM + RR → Grad-CAM / SHAP checks → NSTDB noise.

References

- [1] Q. Xiao et al., "Deep learning-based ECG arrhythmia classification: A systematic review," Applied Sciences, vol. 13, Art. no. 4964, 2023.
- [2] N. N. Kulkarni et al., "Recent advances in deep learning based arrhythmia classification using ECG and PPG signals: A systematic review," Discover Computing, vol. 29, Art. no. 666, 2026.
- [3] J. Beck and A. John, "Explainable AI (XAI) for arrhythmia detection from electrocardiograms," arXiv:2508.17294, 2025.
- [4] B. A. Hassoon et al., "Dual-branch bidirectional attention fusion of temporal and hierarchical representations for robust ECG arrhythmia classification," Expert Systems with Applications, vol. 331, Art. no. 133409, 2026.
- [5] N. Alamatsaz et al., "A lightweight hybrid CNN-LSTM explainable model for ECG-based arrhythmia detection," Biomedical Signal Processing and Control, vol. 90, Art. no. 105884, 2024.
- [6] Sa. I. Ibrahim, "A hybrid deep learning algorithm for ECG-based heart disease classification," Scientific Reports, vol. 16, Art. no. 29911, 2026.
- [7] M. Kachuee, S. Fazeli, and M. Sarrafzadeh, "ECG heartbeat classification: A deep transferable representation," Proc. IEEE ICHI, 2018, pp. 443–444.
- [8] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," Proc. IEEE ICCV, 2017, pp. 618–626.
- [9] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Proc. NeurIPS, 2017, pp. 4765–4774.